SE444: Introduction to Artificial Intelligence

Makeup Quiz: Logic and Bayesian Networks

Fall 2025 – With Solutions

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Instructions

- This quiz contains 10 multiple choice questions.
- Topics: First-Order Logic (FOL), Knowledge Representation, Bayesian Networks, Probabilistic Inference
- Duration: 30 minutes
- Some questions may have multiple correct answers.

Questions and Solutions

Question 1: FOL Translation

Category: First-Order Logic

Statement: "Every hospital has at least one doctor who can treat all patients admitted to that hospital." Which FOL formula best captures this?

A) $\forall h\,(\text{Hospital}(h) \rightarrow \exists d\,(\text{Doctor}(d) \wedge \forall p\,(\text{AdmittedTo}(p,h) \rightarrow \text{CanTreat}(d,p))))$

B) $\exists d\,\forall h\,\forall p\,((\text{Hospital}(h) \wedge \text{AdmittedTo}(p,h)) \rightarrow \text{CanTreat}(d,p))$

C) $\forall h\,\forall p\,(\text{AdmittedTo}(p,h) \rightarrow \exists d\,\text{CanTreat}(d,p))$

D) $\forall d\,\exists h\,(\text{Hospital}(h) \rightarrow \text{WorksAt}(d,h))$

E) $\exists h\,\forall d\,(\text{Doctor}(d) \rightarrow \text{CanTreat}(d,h))$

Answer: A

Rationale: Option A correctly captures the nested quantifiers: for every hospital, there exists a doctor such that for all patients admitted to that hospital, the doctor can treat them. The quantifier scope is critical: $\exists d$ is within the scope of $\forall h$ (allowing different doctors per hospital), and $\forall p$ is within the scope of both.

Question 2: Resolution and CNF

Category: Knowledge Representation

Given the following knowledge base in propositional logic:

1. $(A \land B) \rightarrow C$
2. $C \rightarrow (D \lor E)$
3. $A \land B$
4. $\neg D$

After converting to CNF and applying resolution, what can be inferred?

A) E

B) $\neg E$

C) D

D) $C \land D$

E) The KB is inconsistent

Answer: A

Rationale: Converting to CNF:

- $(A \land B) \rightarrow C$ becomes $\neg A \lor \neg B \lor C$
- $C \rightarrow (D \lor E)$ becomes $\neg C \lor D \lor E$
- $A \land B$ becomes A and B
- $\neg D$ remains $\neg D$

Resolution steps: From A, B, and $\neg A \lor \neg B \lor C$, we get C. From C and $\neg C \lor D \lor E$, we get $D \lor E$. From $D \lor E$ and $\neg D$, we get $\boxed{E}$.

Question 3: Unification and Substitution

Category: First-Order Logic

Given the predicates:

- $P(f(x), g(y), x)$
- $P(f(a), z, a)$

What is the Most General Unifier (MGU)?

A) $\{x/a, z/g(y), y/b\}$

B) $\{x/a, z/g(y)\}$

C) $\{x/a, y/a, z/g(a)\}$

D) $\{x/a, z/g(a)\}$

E) Unification is not possible

Answer: B

Rationale: To unify:

- $f(x)$ must match $f(a) \Rightarrow x = a$
- $g(y)$ must match $z \Rightarrow z = g(y)$
- x must match $a \Rightarrow x = a$ (already unified)

The MGU is $\{x/a, z/g(y)\}$ because it doesn't unnecessarily bind y to a specific value. Option C binds y/a which is more specific than necessary.

Question 4: Bayesian Network Inference

Category: Bayesian Networks

In a Bayesian Network with variables $\{A, B, C, D, E\}$ forming the structure $A \to B \to C$ and $A \to D \to E$, you want to compute $P(E \mid B = b)$. Which variables must be eliminated during Variable Elimination?

A) Only C and D

B) Only A, C, and D

C) Only A and D

D) Only D

E) A, B, C, and D

Answer: C

Rationale: For $P(E \mid B = b)$:

- Query variable: E
- Evidence variable: B
- Hidden variables (to eliminate): A, C, D

However, since C is independent of E given B (d-separation), we only need to eliminate variables on the path from B to E. The path is $B \leftarrow A \to D \to E$, so we must eliminate A and D. Variable C can be ignored as it doesn't affect the query.

Question 5: Quantifier Scope and Meaning

Category: First-Order Logic

Which statement is TRUE about the following two FOL formulas?

1. $\forall x \, \exists y \, \text{Loves}(x, y)$ – "Everyone loves someone"
2. $\exists y \, \forall x \, \text{Loves}(x, y)$ – "There is someone loved by everyone"

A) They are logically equivalent

B) Formula (2) implies Formula (1)

C) Formula (1) implies Formula (2)

D) They are both always false

E) Neither implies the other

Answer: B

Rationale: If there exists someone y loved by everyone (Formula 2), then certainly everyone loves someone (Formula 1) – namely, that particular y. So Formula (2) $\Rightarrow$ Formula (1). However, the converse is not true: everyone loving someone doesn't mean there's one person loved by all. Formula (1) allows each person to love different people.

Question 6: Factor Operations in VE

Category: Bayesian Networks

You have three factors: $f_1(A, B)$, $f_2(B, C)$, and $f_3(C, D)$. You want to eliminate variables in order: B, then C. What is the scope of the final resulting factor after eliminating both?

A) (A, D)

B) (A, B, D)

C) (A, C, D)

D) (B, C)

E) (A)

Answer: A

Rationale:

- **Eliminate B:** Multiply factors containing B: $f_1(A, B) \times f_2(B, C) = f_4(A, B, C)$. Sum out B: $\sum_B f_4(A, B, C) = f_5(A, C)$. Remaining: $f_5(A, C)$ and $f_3(C, D)$.
- **Eliminate C:** Multiply factors containing C: $f_5(A, C) \times f_3(C, D) = f_6(A, C, D)$. Sum out C: $\sum_C f_6(A, C, D) = f_7(A, D)$.

Final scope: $\boxed{(A, D)}$.

Question 7: Bayes' Theorem Application

Category: Bayesian Networks

A rare genetic disorder affects 0.1% of the population. A diagnostic test has 98% sensitivity (true positive rate) and 95% specificity (true negative rate). If a randomly selected person tests positive, what is the approximate probability they have the disorder?

A) About 2%

B) About 16%

C) About 50%

D) About 98%

E) Cannot be determined

Answer: A

Rationale: Using Bayes' theorem:

$$\begin{aligned} P(D \mid +) &= \frac{P(+ \mid D) \cdot P(D)}{P(+ \mid D) \cdot P(D) + P(+ \mid \neg D) \cdot P(\neg D)} \\ &= \frac{0.98 \times 0.001}{0.98 \times 0.001 + 0.05 \times 0.999} \\ &= \frac{0.00098}{0.00098 + 0.04995} \\ &= \frac{0.00098}{0.05093} \approx \boxed{1.92\%} \end{aligned}$$

Key Insight: Despite high test accuracy, the very low prevalence (0.1%) means most positive results are false positives!

Question 8: Skolemization

Category: First-Order Logic

Given the formula: $\forall x \, \exists y \, \forall z \, (P(x, y) \rightarrow Q(y, z))$. Which is the correct Skolemized form?

A) $\forall x \, \forall z \, (P(x, f(x)) \rightarrow Q(f(x), z))$

B) $\forall x \, \forall z \, (P(x, y) \rightarrow Q(y, z))$

C) $\forall x \, \forall z \, (P(x, f(x, z)) \rightarrow Q(f(x, z), z))$

D) $\exists y \, \forall x \, \forall z \, (P(x, y) \rightarrow Q(y, z))$

E) $\forall x \, \forall z \, (P(x, f(z)) \rightarrow Q(f(z), z))$

Answer: A

Rationale: Skolemization replaces existentially quantified variables with Skolem functions. Since $\exists y$ appears within the scope of $\forall x$ but before $\forall z$, the Skolem function for y depends only on x: $y = f(x)$. The $\forall z$ quantifier comes after, so z is not an argument to the Skolem function. Result: $\forall x \, \forall z \, (P(x, f(x)) \rightarrow Q(f(x), z))$.

Question 9: d-Separation

Category: Bayesian Networks

Consider a Bayesian Network: $A \rightarrow B \leftarrow C$ (a V-structure with B as the common effect). Which statement is TRUE?

A) A and C are conditionally independent given B

B) A and C are independent (unconditionally)

C) A and C become dependent when B is observed

D) A and C are always dependent regardless of evidence

E) None of the above

Answer: B and C

Rationale: In a V-structure:

- **Unconditionally:** $A \perp C$ (they are independent without evidence on B)
- **Given B:** A and C become **dependent** (explaining away effect)

Example: A = "Fire", C = "Earthquake", B = "Alarm". Without knowing alarm status, fire and earthquake are independent. But if alarm rings, learning about fire makes earthquake less likely (explaining away).

Accept both B and C as correct answers.

Question 10: Complex Inference with VE

Category: Bayesian Networks

In Variable Elimination, which statement about elimination ordering is TRUE?

A) Different elimination orders can produce different final probability values

B) The optimal elimination order always eliminates variables in alphabetical order

C) A poor elimination order can create exponentially large intermediate factors, drastically increasing computation time

D) Evidence variables should be eliminated first

E) Query variables must be eliminated last

Answer: C

Rationale:

- **A is FALSE:** Different orders always produce the **same** final probability (VE is exact), but with different computational costs.
- **B is FALSE:** Optimal order depends on network structure, not alphabetical order.
- **C is TRUE:** Poor ordering can create huge intermediate factors involving many variables, causing exponential blowup in time/space.
- **D is FALSE:** Evidence variables are **never** eliminated; they are observed.
- **E is FALSE:** Query variables are **never** eliminated; we're computing their probability.

Answer Key Summary

Q#	Topic	Answer	Difficulty
1	FOL Translation	A	HARD
2	Resolution and CNF	A	HARD
3	Unification (MGU)	B	HARD
4	Bayesian Inference (VE)	C	HARD
5	Quantifier Scope	B	EASY
6	Factor Operations	A	HARD
7	Bayes' Theorem	A	HARD
8	Skolemization	A	HARD
9	d-Separation	B, C	MEDIUM
10	VE Elimination Order	C	HARD

Difficulty Distribution:

- Easy: 1 question (Q5)
- Medium: 1 question (Q9)
- Hard: 8 questions (Q1, Q2, Q3, Q4, Q6, Q7, Q8, Q10)

Topic Distribution:

- First-Order Logic / Knowledge Representation: 5 questions (Q1, Q2, Q3, Q5, Q8)
- Bayesian Networks / Probabilistic Inference: 5 questions (Q4, Q6, Q7, Q9, Q10)